Literature Review: Airline Delay Cascades

STA 473, Tools and Techniques for Data Science ·

1. Introduction

When an aircraft lands late, its next flight often leaves late, and the delay can travel through the rest of that aircraft's day. This review covers studies that measure this propagation, mainly those using flight-level on-time records like the US Bureau of Transportation Statistics (BTS) data used here. It ends with the gap that our Cascade Impact Score and PySpark RDD pipeline address.

2. Terminology

A primary delay has its own cause and directly affects one flight. A reactionary (knock-on) delay exists only because an earlier flight was late (EUROCONTROL, 2010). A joint EUROCONTROL and RWTH Aachen study found that 40% to 50% of delays in Europe were reactionary, with less in hub-and-spoke operations than in point-to-point and low-cost ones (EUROCONTROL, 2010).

The BTS late-aircraft delay column is the US counterpart. In our cleaned 2015 data it is never filled for a flight that arrived less than 15 minutes late, which matches the 15-minute threshold in our cascade rule.

3. Aircraft-Level Propagation and Slack

Brueckner, Czerny and Gaggero (2022) define a propagated departure delay as one that occurs when the inbound arrival delay exceeds the next flight's scheduled ground buffer. They look only at a flight and its immediate predecessor, not the whole sequence of flights. Our link rule uses the same buffer idea.

AhmadBeygi, Cohn, Guan and Belobaba (2008) show that schedules built for high aircraft utilization leave little slack to absorb disruption, using data from a hub-and-spoke carrier and a low-fare point-to-point carrier. AhmadBeygi, Cohn and Lapp (2010) build models that reduce propagation by reallocating the slack already in a schedule. These papers support our slack analysis and our what-if buffer simulation.

4. Network-Level Propagation

Fleurquin, Ramasco and Eguiluz (2013) model the US network with 2010 flight data. Delay spreads through aircraft rotation, passenger and crew connections, and airport congestion, and connections were the strongest single mechanism. Pyrgiotis, Malone and Odoni (2013) use a queuing model of the 34 busiest US airports. Both are model-based. Ours works directly from the recorded delays, but it follows aircraft rotation only.

5. Bottleneck Flights

Arikan, Deshpande and Sohoni (2013) model how a root delay propagates through an aircraft's rotation and define a metric for bottleneck flights. This is the idea behind our root flight and Cascade Impact Score. Their model estimates expected propagation, while our score counts the delay minutes that actually passed down each observed chain.

6. Research Gap

In the studies reviewed, this project differs in four ways:

- **Whole chains:** we rebuild each aircraft's full chain from the data, instead of one flight and its predecessor or a fitted model.
- **A damage score:** the Cascade Impact Score gives each flight, airport and route the downstream minutes it caused.
- **Scale:** the pipeline processes about 5.7 million flights with PySpark RDDs, and we will benchmark it against DataFrames.
- **Validation:** we check chains against the late-aircraft column and against synthetic cascades with known ground truth, and then test extra turnaround time in a what-if simulation.

Limitations: our chains follow aircraft rotation only, since BTS data has no crew, passenger or congestion information, and cancelled or diverted flights (about 1.8%) are excluded.

7. References

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